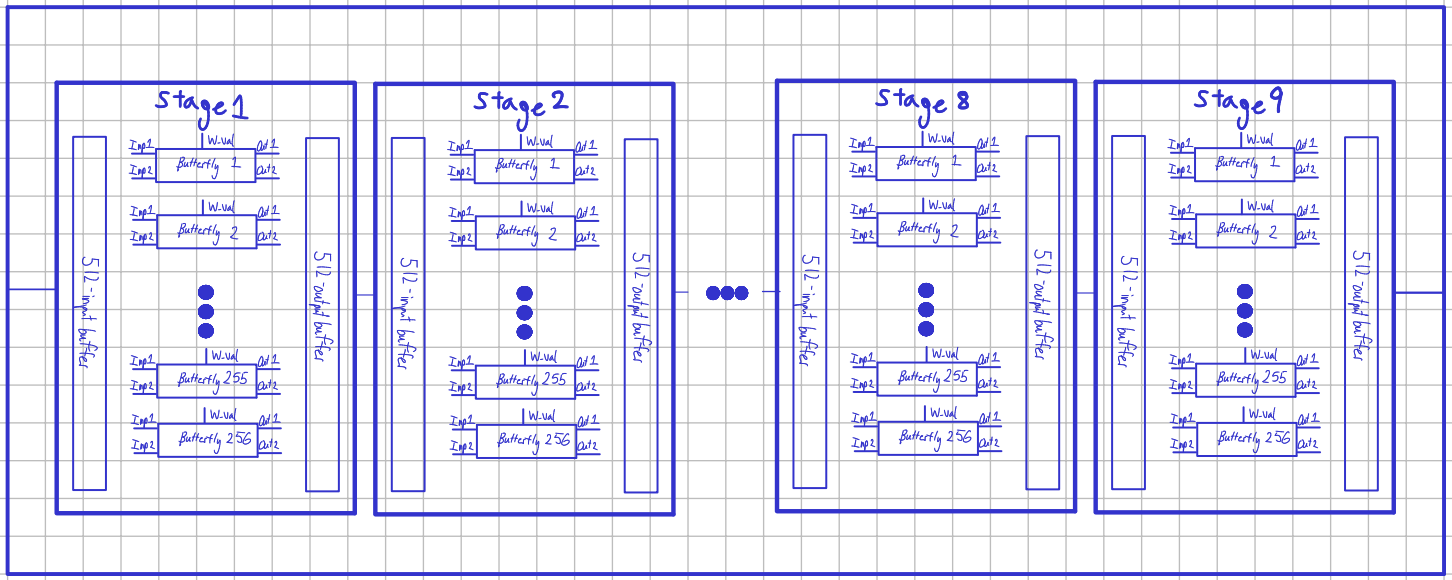
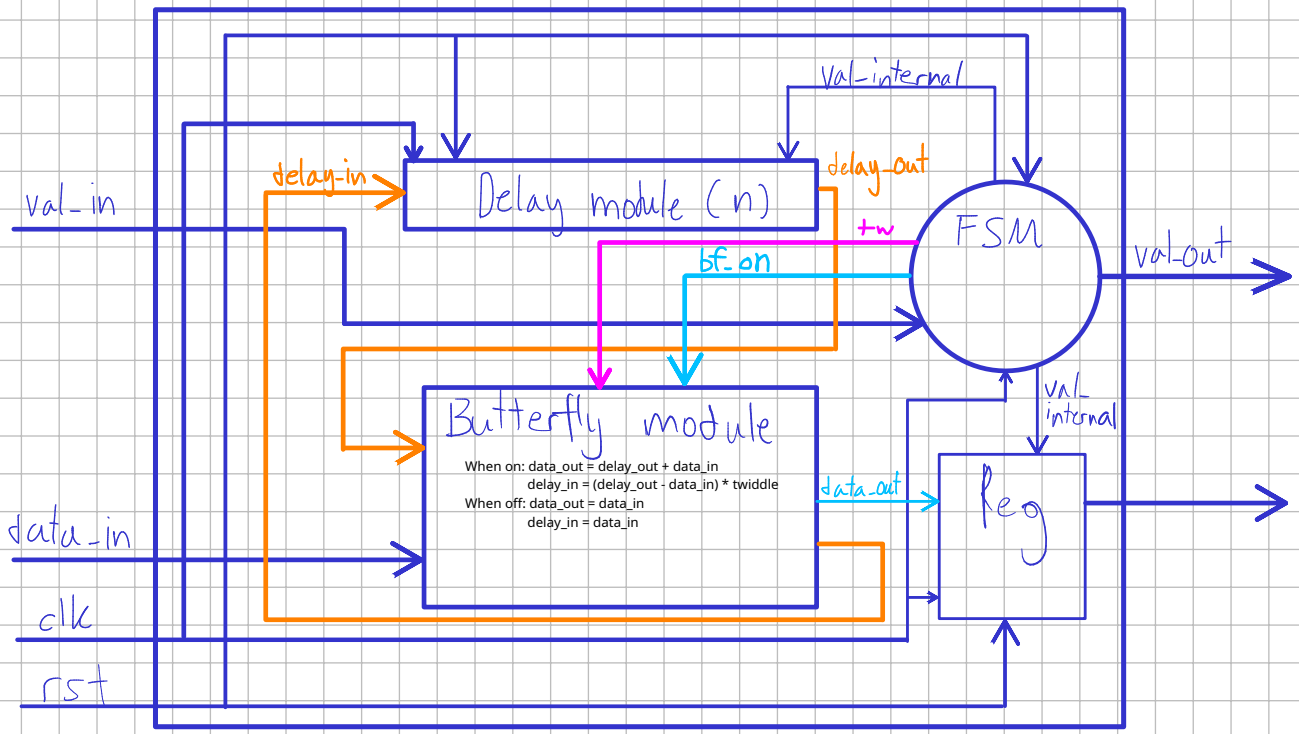
How are we going to make the 512-point FFT RTL model?



The W_val and configuration of how inputs and outputs are made in each butterfly is defined by the current stage number it is inside of (it's going to use parameters to decide the settings of the modules within it).

Previous idea takes too much resources. As a result, we must use conventional SDF/MDC FFT approach to accommodate our FPGA.

SDF Module

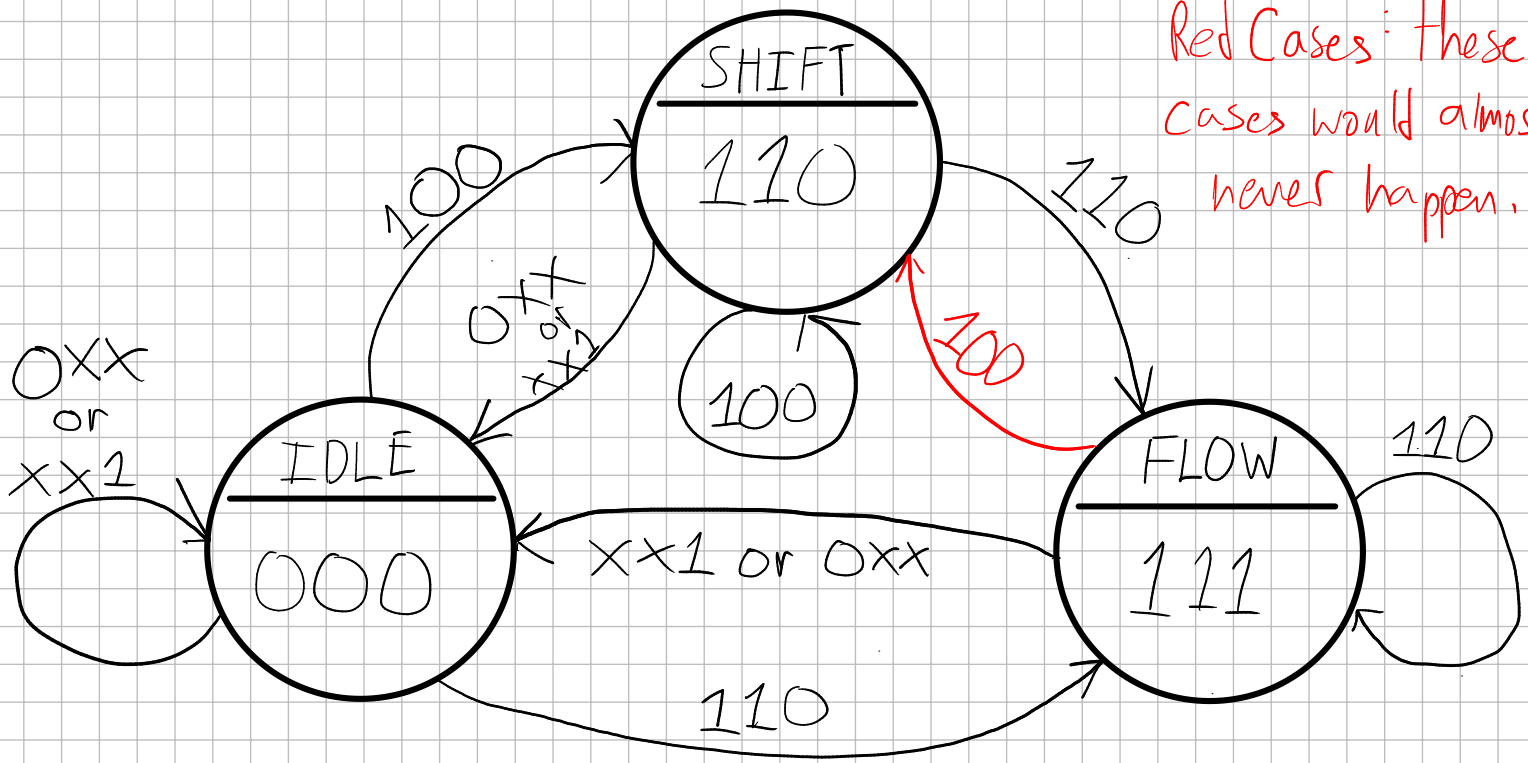


Delay module is shift register of size n, which is dependent on what stage of the FFT this module is being instantiated and the size of the FFT in general (512-point in our case). Equation to define the size can be n = $2^{(\log_2 P - S)}$, such that P = total sample size per FFT (512) and S is the current stage this module is in (1 <= s <= 9)

FSM Diagram of the FSM in the SDF Module

Inputs: {val_in, n_max, rst}

Output: {bf_on, val_internal, val_out}

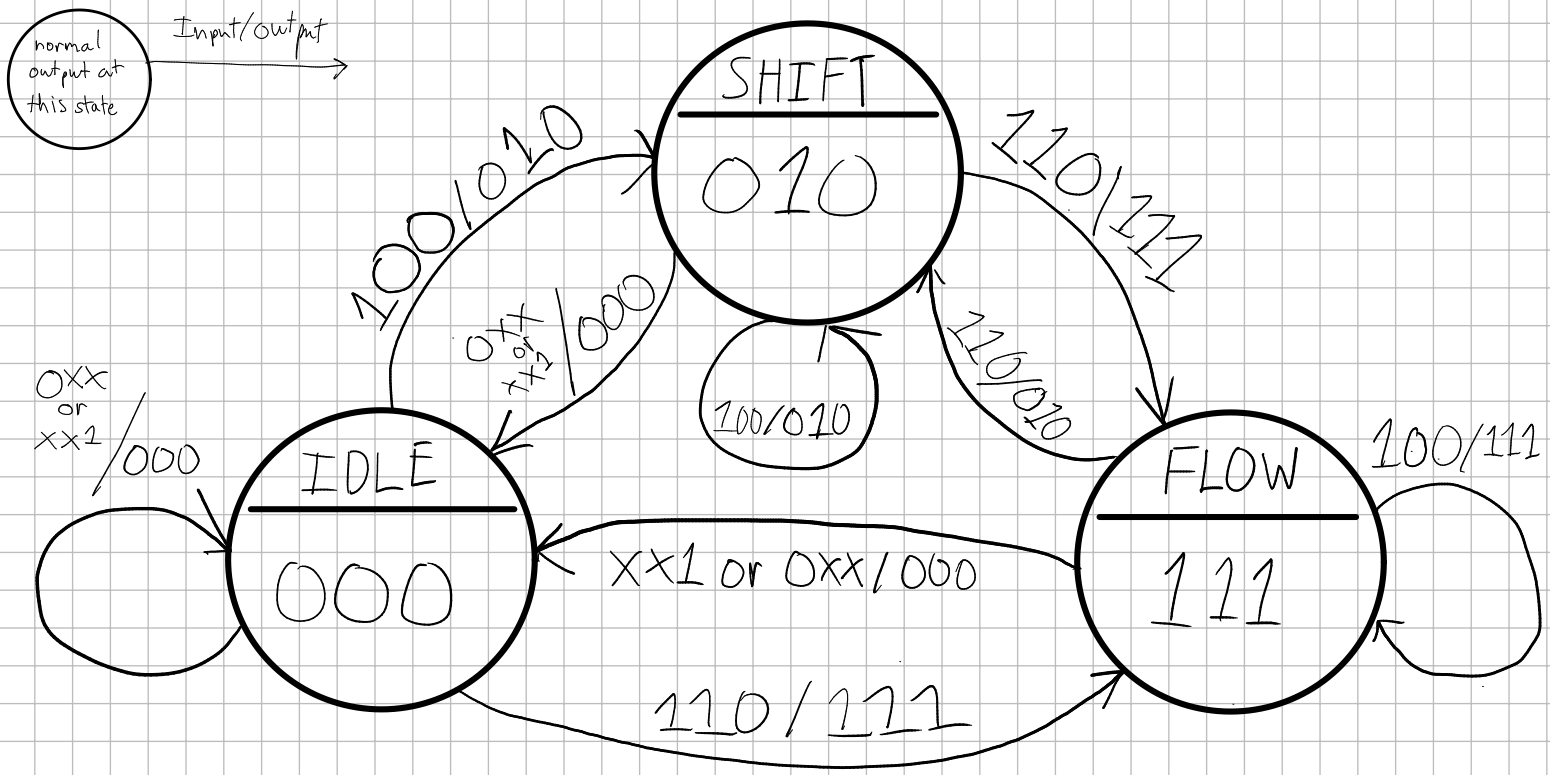


Red Cases: these cases would almost never happen.

FSM Diagram of the FSM in the SDF Module: Attempt 2

Inputs: {val_in, n_max, rst}

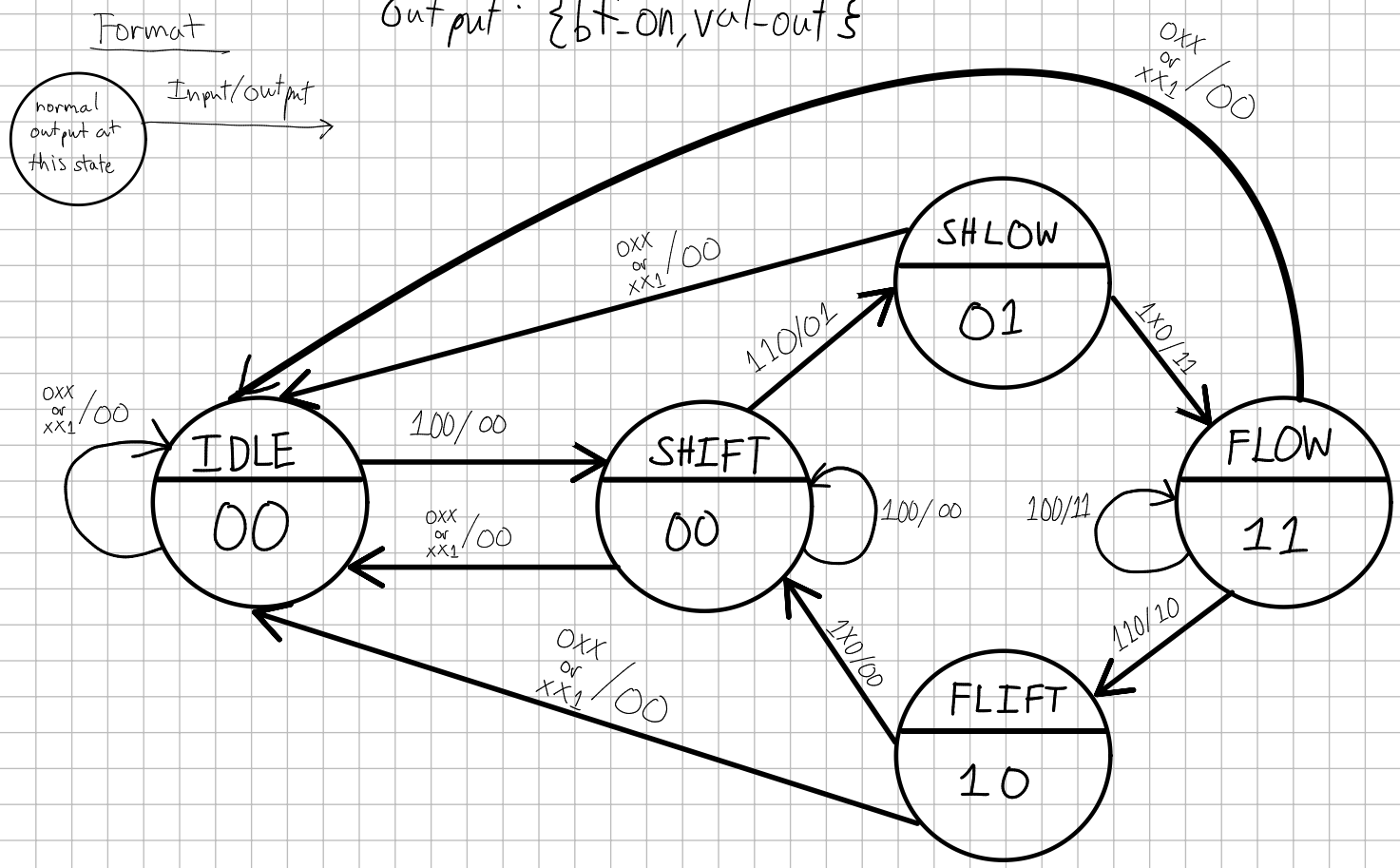
Output: {bf_on, val_internal, val_out}



FSM Diagram of the FSM in the SDF Module: Attempt 3

Inputs: {val_in, n_max, rst}

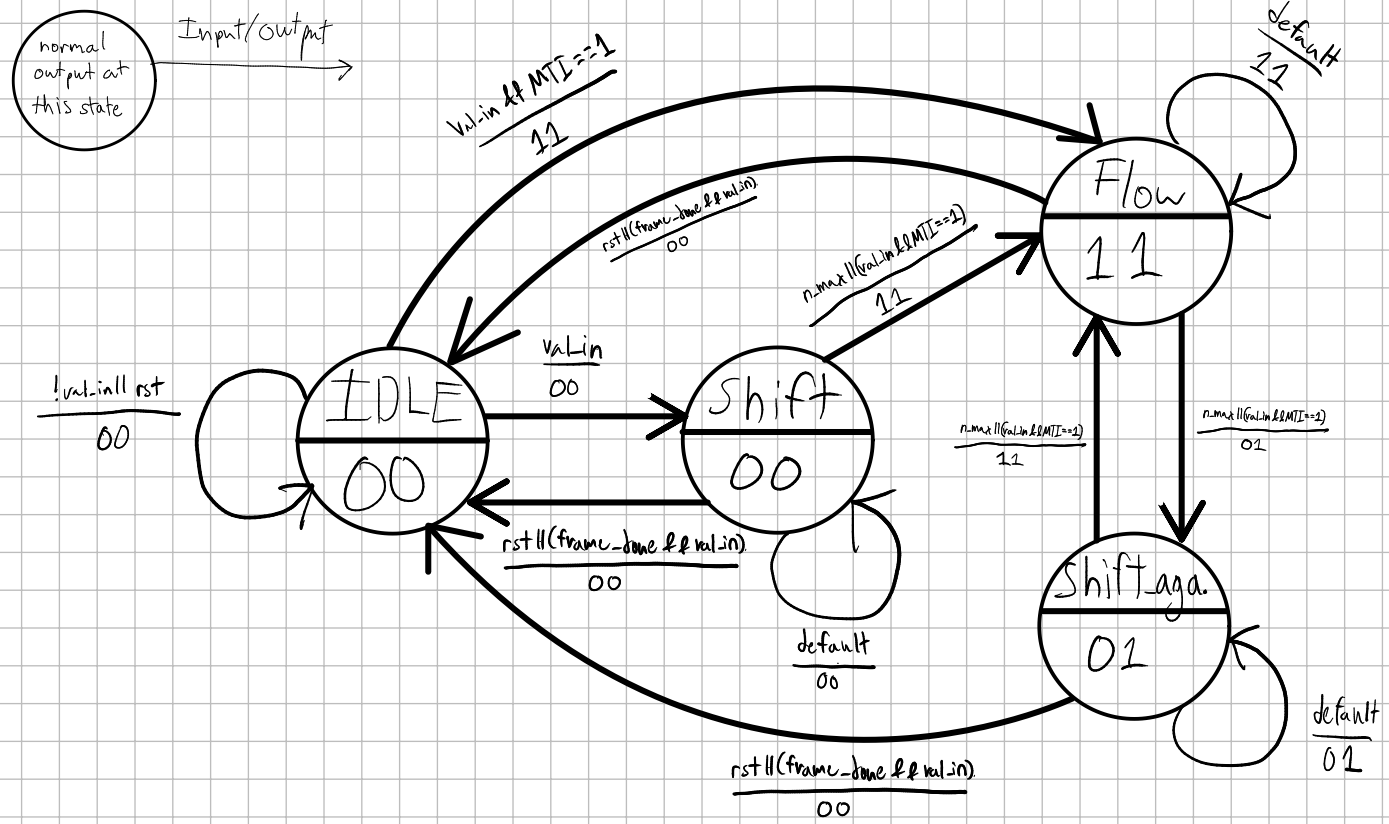
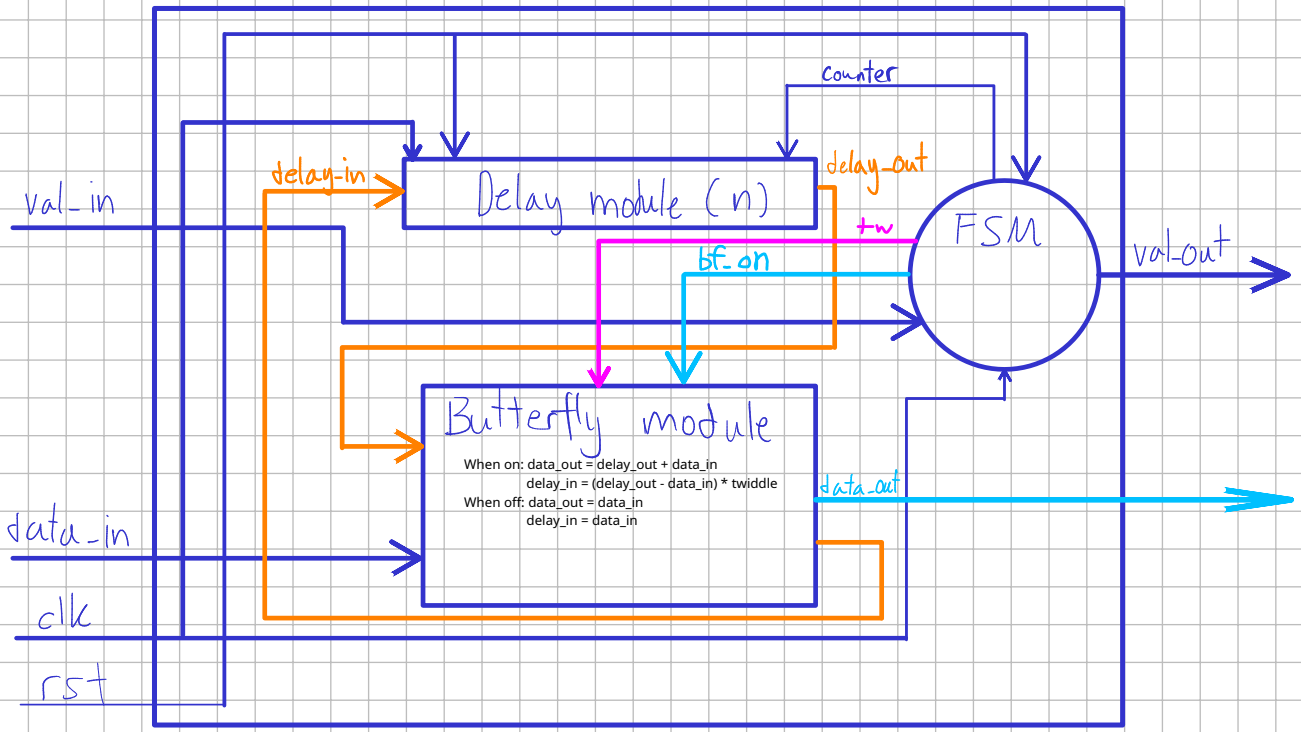
Output: {bf_on, val_out}



FSM Diagram of the FSM in the SDF Module: Attempt 4

Inputs: {val_in, n_max, rst, MTI, frame_done}

Output: {bf_on, val_out}



Corrected SDF Module